Strategy Learner - Optimization Hints and Tips

Optimization Learner Hints and Tips

Overview

The optimization learner is not a stand-alone learner but must be used in conjunction with the implemented Supervised or Reinforcement Learner. You will use the optimizer (which was used earlier in the course) to optimize the lookback window for values indicators. For example, you might find that the best performance is obtained when SMA has lookback value of 13 (e.g., SMA(13)) and RSI has a lookback value of 4 (e.g., RSI(4)).

Additionally, if you are using a Classification learner, you might additionally use optimization to determine the best values for thresholds used to construct your dataset (i.e., YBUY, YSELL).

It will be challenging to use the optimizer to determine a learner's hyperparameters; and doing so may result in a solution that does not meet the assignment's performance requirements.

Hints & Tips

Recall your manual trader: You should have used one or more indicators, then you used a simple set of logical statements to decide on an action. Those logical statements should include statements like “if indicator1 < threshold1 and indicator2 > threshold2” The idea for this version of the project is to use an optimizer to find the values of these thresholds rather than setting them manually. More succinctly, the optimization problem is:

- Given: a set of parameters: threshold[0] to threshold[n-1]
- Find: values for the parameters that maximize Sharpe ratio of the corresponding trading strategy.

Your objective function for the optimizer is thus the Sharpe ratio of the strategy. The logic you use should classify a particular state (set of indicator values) as:

- +1: LONG
- 0: CASH
- -1: SHORT

The X data for each sample (day) are simply the values of your indicators for the stock — you should have 3 to 5 of them. Once you have these values, you can use the classifications to generate a set of transactions which you then assess the Sharpe ratio of.